

Supplementary Material

Reframing preprocessing selection as model-internal calibration in near-infrared spectroscopy

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1 Purpose of the Supplement

This supplement documents the evidence chain behind the main manuscript. It is deliberately more operational than the article: local result sources are listed, claims are mapped to files, deployable and oracle quantities are separated, and reviewer-sensitive weaknesses are made explicit. The goal is to make the AOM study auditable rather than to hide ongoing work behind a single compressed results table.

The supplement uses existing local artifacts only. No additional benchmark was launched to prepare this draft.

For a Talanta submission, the supplement should ultimately become the place where full dataset tables, paired statistical tests, operator-selection frequencies, failure cases, leakage audits and software reproducibility metadata are made explicit. The present version already establishes that structure and marks the remaining experiments required before submission.

2 Notation and Linear-Operator Scope

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a centered spectral matrix and $\mathbf{Y} \in \mathbb{R}^{n \times q}$ a centered response matrix. A strict linear spectral operator $\mathbf{A}_b \in \mathbb{R}^{p \times p}$ acts as

$$\mathbf{X}_b = \mathbf{X}\mathbf{A}_b^\top. \quad (1)$$

The operator bank always contains the identity. This makes standard PLS or standard Ridge available to the selection procedure.

The strict-linear scope includes smoothing filters, derivative filters, finite differences and detrending projections. It does not include sample-adaptive or fitted transformations such as SNV, MSC, EMSC or ASLS. Those transformations are useful in NIRS but must be treated as fold-local pipeline branches. This distinction is central to the paper because the fast AOM algebra is valid only for fixed operators inside a calibration fold.

3 AOM-PLS Derivation

3.1 Covariance identity

For a fixed linear operator $\mathbf{A}_b$, the transformed cross-covariance is

$$\mathbf{X}_b^\top \mathbf{Y} = (\mathbf{X}\mathbf{A}_b^\top)^\top \mathbf{Y} = \mathbf{A}_b \mathbf{X}^\top \mathbf{Y}. \quad (2)$$

This identity permits candidate operators to be evaluated in covariance space for SIMPLS and in adjoint form for NIPALS. It avoids materializing every transformed matrix $\mathbf{X}\mathbf{A}_b^\top$ when the operator can be applied directly to vectors or matrices.

3.2 Original-space coefficients

If a PLS component is estimated in transformed space with direction r_a , the corresponding original-space direction is

$$z_a = \mathbf{A}_b^\top r_a. \quad (3)$$

Collecting k such directions gives $\mathbf{Z} = [z_1, \dots, z_k]$ and scores $\mathbf{T} = \mathbf{X}\mathbf{Z}$. The final coefficient matrix is

$$\mathbf{B} = \mathbf{Z}(\mathbf{P}^\top \mathbf{Z})^+ \mathbf{Q}^\top. \quad (4)$$

Thus AOM-PLS remains a single linear calibration model on the original wavelength grid.

4 AOM-Ridge Derivation

For centered $\mathbf{X}_c$ and $\mathbf{Y}_c$, Ridge regression can be solved in the dual. A single operator gives

$$\mathbf{Z}_b = \mathbf{X}_c \mathbf{A}_b^\top, \quad (5)$$

$$\mathbf{K}_b = \mathbf{Z}_b \mathbf{Z}_b^\top = \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top, \quad (6)$$

$$\mathbf{C} = (\mathbf{K}_b + \alpha \mathbf{I}_n)^{-1} \mathbf{Y}_c. \quad (7)$$

The original-space coefficient matrix is

$$\beta_b = \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top \mathbf{C}. \quad (8)$$

For a superblock or weighted operator mixture,

$$\mathbf{K} = \sum_{b=1}^m s_b^2 \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top, \quad \mathbf{M} = \sum_{b=1}^m s_b^2 \mathbf{A}_b^\top \mathbf{A}_b, \quad (9)$$

and the original-space coefficient matrix is

$$\beta = \mathbf{M} \mathbf{X}_c^\top (\mathbf{K} + \alpha \mathbf{I}_n)^{-1} \mathbf{Y}_c. \quad (10)$$

5 Algorithmic Sketches

5.1 Global AOM-PLS

1. Build a compact operator bank containing identity and a small set of smoothing, derivative and detrending operators.
2. For each candidate number of components and each operator, evaluate calibration-fold performance.
3. Select the operator and component count with the best calibration-fold criterion.
4. Refit on the complete calibration set.
5. Store selected operator, component count, original-space coefficients, centering constants and fold-level diagnostics.

5.2 AOM-Ridge

1. Build operator-induced kernels $\mathbf{K}_b = \mathbf{X}_c \mathbf{A}_b^\top \mathbf{A}_b \mathbf{X}_c^\top$ inside the calibration fold.
2. Select α and operator geometry by nested calibration-fold validation.
3. Solve the dual Ridge system.
4. Recover original-space coefficients.
5. Refit on the complete calibration set and evaluate once on the external test set.

6 Operator and Branch Catalogue

7 Benchmark Protocol Ledger

Table 1: Strict operators and branch preprocessors considered in the AOM study.

Group	Examples	Treatment in the paper
Identity	I	Always included; recovers the standard model.
Smoothing	Savitzky–Golay smoothing	Strict linear operator.
Derivatives	SG first and second derivatives, finite difference	Strict linear operator.
Detrending	degree-1 and degree-2 trend projections	Strict linear operator when fitted inside folds.
Scatter correction	SNV, MSC, EMSC	Fold-local branch preprocessing, not a fixed operator.
Baseline correction	ASLS and related smoothers	Fold-local branch preprocessing, not a fixed operator.
Orthogonal correction	OSC-style branches	Fold-local branch; not part of the strict AOM algebra.

Table 2: Local evidence map used to prepare the manuscript.

Artifact	Content	Use in the paper
<code>bench/AOM_v0/Summary.md</code>	57-dataset AOM-PLS summary; compact ASLS+AOM CV-5 champion.	Primary AOM-PLS headline source.
<code>bench/AOM_v0/publication/tables/relative_rmsep_per_variant.csv</code>	Relative RMSEP and fit-time summaries for AOM-PLS variants.	Source for median RMSEP/PLS, wins and fit time.
<code>bench/AOM_v0/docs/AOMPLS_MATH_SPEC.md</code>	AOM-PLS operator algebra, NIPALS/SIMPLS conventions.	Method verification and notation.
<code>bench/AOM_v0/docs/AOMPLS_VALIDATION.md</code>	Equivalence tests: identity, fixed operator, NumPy/Torch parity.	Implementation validation.
<code>bench/AOM_v0/Ridge/docs/AOM_RIDGE_MATH_SPEC.md</code>	Dual AOM-Ridge derivation and branch warnings.	Ridge method section.
<code>bench/AOM_v0/Ridge/publication/tables/table_per_method_summary.tex</code>	Deployable AOM-Ridge variants on 52 datasets.	Conservative deployable Ridge claim.
<code>bench/AOM_v0/Ridge/publication/tables/table_summary.tex</code>	Oracle envelope against reference baselines.	Upper-bound Ridge claim only.
<code>bench/AOM_v0/Ridge/docs/HEADLINE_SPXY3_NESTED_AUDIT.md</code>	Nested-selection audit for headline Ridge variants.	Leakage-risk discussion.
<code>bench/tabpfn_paper/article/main.tex</code>	Reference benchmark protocol, preprocessing search budgets and splits.	Baseline protocol context; model name not emphasized in the article.
<code>bench/benchmark_master_results.csv</code>	Unified registry of many variants and maturity levels.	Exploration registry, not a direct source of all headline statistics.

8 Claim Ledger

Table 3: Main claims and their current evidence status.

Claim	Status	Dataset count	Source / caveat
ASLS + compact AOM-PLS CV-5 improves over PLS with median RMSEP/PLS 0.960.	Deployable AOM-PLS benchmark result.	57	<code>bench/AOM_v0/Summary.md</code> ; needs final cohort manifest for journal submission.
AOM-PLS compact CV-5 wins against PLS on 42 datasets.	Deployable AOM-PLS benchmark result.	57	Same source; no multi-seed uncertainty yet.
AOM-PLS compact CV-5 median fit time is approximately 1.36 s.	Measured in local benchmark table.	57	Hardware-specific; should be remeasured on the final machine.
AOM-Ridge Blender improves over tuned Ridge by median 2.22%.	Deployable AOM-Ridge table result.	52	<code>table_per_method_summary.tex</code> ; single-seed cohort.
AOM-Ridge Blender wins 35/52 against tuned Ridge.	Deployable AOM-Ridge table result.	52	Nested-audit evidence exists, but full multi-seed promotion is pending.
AOM-Ridge oracle improves over tuned Ridge by median 4.73%.	Oracle envelope only.	52	<code>table_summary.tex</code> ; not a deployable method claim.
AOM-Ridge oracle wins 45/52 against tuned Ridge.	Oracle envelope only.	52	Use only as headroom / upper-bound evidence.
Large preprocessing searches evaluate thousands of fold-level candidates.	Reference protocol fact.	per dataset	PLS/Ridge approximately 5400, CNN approximately 2700, CatBoost approximately 90.

9 AOM-PLS Variant Summary

Table 4: Selected AOM-PLS variants from the existing 57-dataset summary.

Variant	Datasets	Median RMSEP/PLS	Wins	Median fit (s)
ASLS-AOM compact CV-5	57	0.960	42	1.36
ASLS-AOM response-dedup CV-3	57	0.964	37	4.57
ASLS-AOM family-pruned CV-3	57	0.964	38	1.60
ASLS-AOM compact repeated CV-3	57	0.975	39	2.21
ASLS-AOM compact CV-3	57	0.979	38	0.91

Table 5: Deployable AOM-Ridge variants. These are the appropriate primary Ridge-family claims before multi-seed promotion; full local identifiers are in the source table.

Variant	Wins / N	Win-rate (%)	Mean rank	Median Δ vs Ridge (%)
Blender headline SPXY3	35/52	67.3	3.56	-2.22
AutoSelect headline SPXY3	27/52	51.9	4.61	-0.61
Global compact none	29/52	55.8	4.79	-0.63

Table 6: AOM-Ridge oracle envelope. These values describe headroom and must not be worded as deployable practitioner performance.

Baseline	Median Δ (%)	Mean Δ capped (%)	Win-rate (%)	Wins / N
Ridge	-4.73	-12.37	86.5	45/52
PLS	-7.73	-14.66	90.4	47/52
CNN	-12.56	-16.08	74.5	35/47
CatBoost	-13.27	-19.77	73.1	38/52

10 AOM-Ridge Result Separation

11 Paired Statistical Evidence

Table 7: Current paired statistical evidence for the two primary deployable claims. These values were computed from existing result files and should be regenerated after the final Talanta cohort manifest is frozen.

Comparison	N	Median effect	95% bootstrap CI	Wins	Wilcoxon p_{Holm}
AOM-PLS compact CV-5 vs PLS	57	RMSEP ratio 0.960	0.926–0.990	42/57	1.48×10^{-4}
AOM-Ridge Blender vs tuned Ridge	52	Δ RMSEP -2.22%	-3.66 to -0.83%	35/52	8.05×10^{-3}

12 Leakage and Audit Controls

13 Additional Experiments Required for a Journal Submission

Table 9: Experiment backlog.

Priority	Experiment	Purpose
P0	Freeze one regression cohort manifest.	Remove denominator drift across 52/57/58/61 dataset statements.
P0	Regenerate headline tables from one manifest.	Ensure PLS, Ridge, CatBoost, CNN, AOM-PLS and AOM-Ridge are compared on a common subset.
P0	Separate deployable and oracle tables.	Prevent reviewers from interpreting oracle envelopes as methods.
P0	Add exclusion table.	Document QUARTZ-style degenerate denominators, missing CNN rows and compute skips.

Priority	Experiment	Purpose
P1	Full multi-seed rerun.	Quantify selection variance and win-rate uncertainty.
P1	Paired statistical tests.	Add Wilcoxon/Friedman/Nemenyi or equivalent tests on the common subset.
P1	Strict-linear vs branch ablation.	Separate gains from AOM itself and gains from ASLS/SNV/MSB branches.
P1	Operator-frequency table.	Show which spectral operations are actually selected.
P1	Runtime scaling study.	Quantify cost versus n , p , bank size and branch count.
P1	Prediction artifact export.	Enable residual plots, bias, RPD/RPIQ, and independent audit.
P1	Software release validation.	Validate the clean-room C++ AOM-PLS implementation, R bindings, Python package AOM-PLS, and planned pip AOM-PLS/AOM-Ridge packages against the reference implementation.

14 Reproducibility Checklist

1. Record repository commit hash, Python version and package versions.
2. Store the cohort manifest used for every table.
3. Store split indices for every dataset.
4. Store fold predictions and final test predictions.
5. Store selected operators, component counts, α values and branch preprocessing choices.
6. Store runtime and hardware metadata.
7. Regenerate tables from scripts rather than manually editing manuscript tables.

15 Recommended Positioning

The safest wording for the current draft is: AOM is a general linear-operator framework; AOM-PLS has the strongest deployable full-cohort evidence; AOM-Ridge is a mathematically consistent and promising extension with deployable evidence and a stronger oracle envelope, but requires full multi-seed promotion before being described as a production default. This position is scientifically stronger than claiming universal benchmark dominance.

References

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Table 8: Leakage controls that must be preserved in any final rerun.

Control	Requirement
External test set	Never used for operator, branch, component or α selection.
Original splits	Preserved when supplied by the dataset source.
Generated splits	Deterministic SPXY or stratified variants, with split type logged.
Branch preprocessing	Fitted inside each calibration fold only.
Operator selection	Performed using calibration-fold validation only.
Final refit	Refit selected configuration on the full calibration set, then evaluate once on test.
Oracle tables	Clearly labeled as retrospective upper bounds.
Skipped datasets	Logged with inclusion/exclusion reason.

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